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Inference

Note that the graph only tells us what the sales team had forecast for March and April (16 and 20 units sold, respectively), not the actual units sold.

1. This statement tells us only the actual number of units sold in March but nothing about the actual number of units sold in April. If the actual sales varied from the forecast sales in a random way from month to month, then it is possible that March's sales were less than, more than, or the same as April's sales; NOT sufficient.
2. Because this statement tells us that there is an unchanging linear relationship between the forecast and the actual number of units of Product Z sold through the months given in the graph, we know that for any month X given in the graph, if another month Y was forecast to have lower sales than X, then the actual sales of Y were lower than the sales of X. Therefore, since the forecast for units sold for March was lower than the forecast for units sold for April, then the actual sales for March were lower than the actual sales for April; SUFFICIENT.

The correct answer is B;statement 2 alone is sufficient.

•

Chris was in an electronic appliances store to buy an LED computer monitor. Chris's purchase criteria included screen size, which must be at least 32 inches, and price, which must not exceed $400. A friend made two recommendations: Monitor A and Monitor B. Chris liked both monitors. Did Chris more likely buy Monitor A, Monitor B, or neither?

1. Monitor A's screen size was more than 32 inches, and it cost less than Monitor B.
2. Monitor B cost $390, and its screen size was less than that of Monitor A.

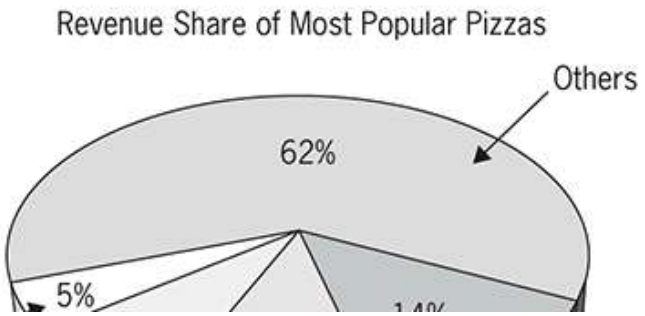
Inference

Chris is in a store shopping for an LED computer monitor that must meet at least two criteria—the monitor must have a screen size of at least 32 inches and cost no more than $400. Note that we are not told whether these are Chris's only two criteria; there may be others. Chris likes both Monitor A and Monitor B, so presumably they both meet his criteria regarding screen size and price.

1. From this statement we can be certain that Monitor A's screen size is sufficiently large, but we cannot infer from it whether Monitor A meets Chris's price requirement. We cannot infer anything about Monitor B's screen size, and it may or may not be the case that its screen size is enough larger than Monitor A's that Chris would be willing to pay more for it, assuming neither monitor is more than $400; NOT sufficient.
2. From this statement we can conclude that Monitor B meets Chris's price criterion, but it does not allow us to infer anything about Monitor A's price—though it does inform us that Monitor B has a smaller screen size than Monitor A. It does not, however, give us information sufficient to infer whether either of the monitors meets his screen size requirement; NOT sufficient.

It might seem that from (1) and (2) taken together, we can infer that Chris would prefer Monitor A to Monitor B and would therefore choose to purchase Monitor A. After all, we can infer from those statements together that Monitor A meets both the screen size requirement and the price requirement (because we can infer that Monitor A costs less than $390) and that Monitor A is both larger and less expensive than Monitor B. But remember that there may be other criteria that Chris may use for assessing which monitor to purchase besides the two criteria mentioned, which might make Monitor B preferable to Monitor A. Furthermore, none of the given information confirms that Monitor A or Monitor B are LED monitors, the kind we are told Chris desires. Furthermore, there may be a third monitor that Chris would choose over either of these two, in which case Chris might be likely to purchase neither. Therefore, (1) and (2) taken together are not sufficient.

The correct answer is E; both statements together are still not sufficient.



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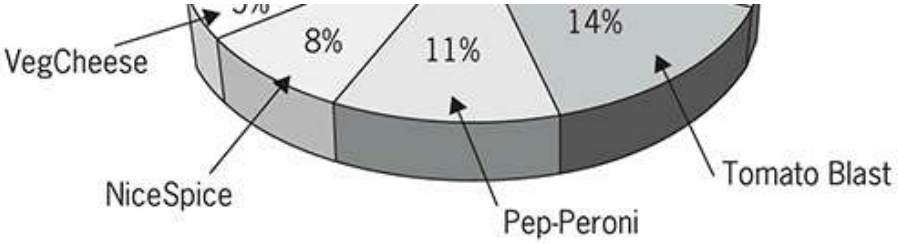

•

A certain pizzeria has more than fifty types of pizza on its menu. The chart above shows the revenue share, in the pizzeria's total revenue in the last month, of its four most popular pizza types. Each pizza type has only a single size and a single price.

Which was their best-selling pizza last month in terms of number of pizzas sold?

1. The number of customers who ordered at least one Pep-Peroni pizza last month was greater than the number of customers who ordered at least one of any other pizza type last month.
2. The lowest-priced pizza type on the pizzeria's menu is Tomato Blast, followed by VegCheese.

Inference

The graph shows the revenue share of the pizzeria's total revenue in the last month of its four most popular types. Each type of pizza has only a single size and a single price, but note that nothing in the initially given information indicates how the prices of the various types of pizza relate to one another—for instance, do all the types have the same price or not, and if not, which type is most expensive. Therefore, even though the graph shows that revenue from sales of Tomato Blast made up the greatest share of the total revenue, it is possible that Tomato Blast was not the best-selling pizza last month in terms of number of pizzas sold.

1. This statement does not allow us to answer the question, because it does not tell us whether the average number of Pep-Peroni pizzas ordered per customer was greater than, less than, or the same as the average number of other types. Therefore, though it is possible that Pep-Peroni was the best-selling pizza last month (in which case Pep-Peroni pizzas must be less expensive than Tomato Blast pizzas), it might not be, if the average number of Pep-Peroni pizzas ordered by customers who bought at least one was less than the average number of pizzas of some other type ordered by customers who bought that type; NOT sufficient.
2. If Tomato Blast is the lowest-priced pizza type, and yet (as the graph shows) revenue from its sales made up the largest share of total revenue, clearly more Tomato Blast pizzas were sold than any other type of pizza; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

In any twenty-four-hour period, a certain faulty clock gains anywhere from one second to a maximum of three minutes. The clock did not stop or lose time since noon the day before yesterday. At exactly noon today, did the clock show the correct time to an accuracy of five minutes?

1. The clock was set to the correct time the day before yesterday at exactly noon.
2. At exactly noon yesterday, the clock was set back by one minute.

Inference

The clock in question gains anywhere from one second to three minutes in any 24-hour period. Given that we know that the clock never stopped or lost time since noon the day before yesterday, we can answer the question if we could know whether the clock was any more than 5 minutes and one second slow at noon yesterday or any more than two minutes fast at noon yesterday.

1. From this statement we only know that the clock could be as much as six minutes fast—in which case the answer to the question is “No.” But nothing rules out that the clock gained only, for instance, two seconds, or that at some point in the past 48 hours someone manually changed the time on the clock; NOT sufficient.
2. Without the information given in (1), (2) does not help determine an answer to the question, because we do not know what time was on the clock immediately prior to its being reset at noon yesterday. If, for instance, the clock was one minute fast just before noon yesterday, then at most the clock would be three minutes fast at noon today, and the answer would be “Yes.” But if the clock at just before noon yesterday was, for instance, four

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minutes fast, then the clock at noon today could have been as much as six minutes fast, in which case that answer would be "No"; NOT sufficient.

From (1) and (2), we can infer that, as of just before noon yesterday, the clock was at least one second fast and at most three minutes fast. Therefore, when the clock was set back one minute at noon, the clock could have been anywhere from fifty-nine seconds slow to two minutes fast. Therefore, because the clock would have gained a minimum of one second and a maximum of three minutes between noon yesterday and noon today, we can infer that at most the clock was fifty-eight seconds slow and at most five minutes fast at noon today. Given that range, we can infer that at noon today, the clock showed the correct time to an accuracy of five minutes, and thus the answer to the question is "Yes," which makes both statements together sufficient.

The correct answer is C; both statements together are sufficient.

•

The couple decide one day that they will definitely go to the beach the next evening if it is sunny without rain at the time they plan to go, and that they will definitely not go if such is not the case. The next morning, the hour-by-hour weather forecast says that in the evening, it will be sunny without rain. Will the couple go to the beach that evening?

1. It is raining on the morning and in the afternoon of the day of their planned beach outing.
2. The couple always believe that the weather forecast is likely to be accurate.

Inference

A couple plans to go to the beach the next evening if and only if it is sunny and not raining at the time they plan to go. The hour-by-hour forecast says that it will be sunny without rain on that next evening.

1. We are told that the couple will go if and only if it is not raining *at the time they plan to go*, which is that next evening, not the morning or afternoon. Therefore, the information in this statement is irrelevant; NOT sufficient.
2. The couple may always believe that the forecast is likely to be accurate, and maybe it is *likely* to be accurate. But that does not mean that it will be accurate this time. Perhaps it will turn out to be one of the rare times when the forecast is inaccurate. The stimulus says that the couple will go if, but only if, it actually is sunny without rain *at the time they plan to go*, not if they have believed on the basis of an earlier forecast that it will not be raining; NOT sufficient.

We cannot infer from (1) and (2) taken together whether or not it is actually sunny and not raining at the time they plan to go to the beach that evening. Therefore, (1) and (2) taken together are not sufficient.

The correct answer is E; both statements together are still not sufficient.

•

A small library has books on twenty different subjects including, most prominently, history. One afternoon, a librarian arbitrarily picks up 100 books for reshelving from among the books returned that day. All the books returned that day were borrowed during the past three weeks. Are fewer than 20 percent of the books the librarian picks up on history?

1. Fewer than 20 percent of the books in the library are on history.
2. During the past three weeks, exactly 15 books on history were borrowed from the library.

Inference

Are fewer than 20 percent of the 100 books that the librarian has arbitrarily picked up to reshelve—all of which had been returned that day—on history? Note that all the books returned that day had been borrowed during the past three weeks.

1. Nothing in the stimulus or (1) tells us whether the distribution of subject matter of the books that were returned that day was representative of the distribution of subject matter of the library in general; NOT sufficient.
2. If during the past three weeks exactly 15 books on history were borrowed from the library, as this statement indicates, and all books returned to the library that day had been borrowed during the past three weeks, it follows that at most 15—and perhaps even none—of the 100 books that the librarian picked up to reshelve were on history. Therefore, no more than 15 percent of those books are on history; SUFFICIENT.

The correct answer is B;statement 2 alone is sufficient.

Topic	Alex	Beika	Carlos	Deepa
History	X	X	✓	X

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History	X	X	X	X
Humor	X	✓	X	✓
Language	X	X	✓	✓
Philosophy	✓	✓	✓	X
Science Fiction	✓	X	✓	✓

•

Four book lovers have recently met and are not sure of the topics that interest the others. The table above indicates the topics that interest each of them with check marks and those that do not with cross marks (Xs).

One of these persons will gift another of these persons two books, each on a different one of the listed topics. Will at least one of the two books fall under the recipient's broad topic of interest?

1. One topic is science fiction and the other is humor.
2. The intended recipient is Carlos.

Inference

The chart lists four book lovers and the book topics that interest them. None of these book lovers knows what topics interest the others, but one of the book lovers gifts another one of them two books, each on a different one of the topics.?

1. One of the two topics is science fiction and the other is humor. Each of the potential recipients is interested in science fiction except for Beika, but Beika is interested in humor. Therefore, no matter which book lover is the recipient, at least one of the two books is on one of the potential recipient's topics of interest; SUFFICIENT.
2. This tells us that Carlos is the intended recipient, and the chart shows that Carlos is interested in all the topics except for humor. So even if one of the two topics is humor, whatever the other topic is, it will be among the recipient's—i.e., Carlos's—topics of interest; SUFFICIENT.

The correct answer is D;each statement alone is sufficient.

•

In a gas sample, there are, rounded to the nearest order of magnitude (that is, to the nearest power of 10), approximately 10^{21} molecules of H_2 and also 10^{21} molecules of O_2. What is the combined number of H_2 and O_2 molecules in the gas sample, rounded to the nearest order of magnitude?

1. The number of H_2 molecules and the number of O_2 molecules are each less than 3×10^{21}.
2. The number of H_2 molecules is more than twice the number of O_2 molecules.

Arithmetic Exponents

We are given that the number of H_2 molecules is greater than 5×10^{20} and the number of O_2 molecules is greater than 5×10^{20}. Therefore, the combined number of molecules is greater than 1×10^{21}, and hence the combined number of molecules is greater than 10^{20} when rounded to the nearest order of magnitude. We are to determine the combined number of molecules when rounded to the nearest order of magnitude.

1. We are given that each of the numbers of H_2 molecules and O_2 molecules is less than 3×10^{21}. It is possible that the combined number of molecules is 10^{21} when rounded to the nearest order of magnitude. For example, if there are 2×10^{21} molecules of each type, then the combined number of molecules is 4×10^{21}, which when rounded to the nearest order of magnitude is 10^{21}. On the other hand, it is possible that the combined number of molecules is 10^{22} when rounded to the nearest order of magnitude. For example, if there are 2.8×10^{21} molecules of each type, then the combined number of molecules is 5.6×10^{21}, which when rounded to the nearest order of magnitude is 10^{22}; NOT sufficient.
2. We are given that the number of H_2 molecules is more than twice the number of O_2 molecules. It is possible that the combined number of molecules is 10^{21} when rounded to the nearest order of magnitude. For example, if there are 2×10^{21} molecules of H_2 and 6×10^{20} molecules of O_2 (note that each rounded to the nearest order of magnitude is 10^{21}), then the number of H_2 molecules is more than twice the number of O_2 molecules, and the

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magnitude is 10^{21}), then the number of H_2 molecules is more than twice the number of O_2 molecules, and the combined number of molecules is 2.6×10^{21}, which when rounded to the nearest order of magnitude is 10^{21}. On the other hand, it is possible that the combined number of molecules is 10^{22} when rounded to the nearest order of magnitude. For example, if there are 4.5×10^{21} molecules of H_2 and 2×10^{21} molecules of O_2 (note that each rounded to the nearest order of magnitude is 10^{21}), then the number of H_2 molecules is more than twice the number of O_2 molecules, and the combined number of molecules is 6.5×10^{21}, which when rounded to the nearest order of magnitude is 10^{22}; NOT sufficient.

Taking (1) and (2) together, from (1) we know the number of H_2 molecules is less than 3×10^{21}, and from (2) we know the number of O_2 molecules is less than 1.5×10^{21}. Therefore, the combined number of molecules is less than 4.5×10^{21}, which is less than 10^{22} when rounded to the nearest order of magnitude. We also know from above (before (1) or (2) is considered) that the combined number of molecules is greater than 10^{20} when rounded to the nearest order of magnitude. Since the combined number of molecules, when rounded to the nearest order of magnitude, is less than 10^{22} and greater than 10^{20}, it follows that the combined number of molecules is 10^{21} when rounded to the nearest order of magnitude.

The correct answer is C; both statements together are sufficient.

•

On Jane's credit card account, the average daily balance for a 30-day billing cycle is the average (arithmetic mean) of the daily balances at the end of each of the 30 days. At the beginning of a certain 30-day billing cycle, Jane's credit card account had a balance of $600. Jane made a payment of $300 on the account during the billing cycle. If no other amounts were added to or subtracted from the account during the billing cycle, what was the average daily balance on Jane's account for the billing cycle?

1. Jane's payment was credited on the 21st day of the billing cycle.
2. The average daily balance through the 25th day of the billing cycle was $540.

Algebra Statistics

1. It is given that the $300 payment was credited on the 21st day of the billing cycle. Thus, the 30 daily balances consisted of 20 daily balances of $600 followed by 10 daily balances of $300, and hence the average daily balance for the 30-day billing cycle was $\frac{(20)(\$600)+(10)(\$300)}{30} = \left(20\right)\left(\$20\right) + \left(10\right)\left(\$10\right) = \$500$; SUFFICIENT.
2. It is given that the average daily balance through the 25th day of the billing cycle was $540. If n is the number of days of the billing cycle before the $300 payment was credited, then the 25 daily balances consisted of n daily balances of $600 followed by $(25-n)$ daily balances of $300. Since the average of these 25 daily balances was $540, it follows that $\frac{(n)(\$600)+(25-n)(\$300)}{25} = 540$, which can be solved to obtain $n = 20$. Therefore, the $300 payment was credited on the 21st day of the billing cycle, and the same calculation used to evaluate (1) above shows the average daily balance for the 30-day billing cycle was $500; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

The annual rent collected by a corporation from a certain building was x percent more in 1998 than in 1997 and y percent less in 1999 than in 1998. Was the annual rent collected by the corporation from the building more in 1999 than in 1997?

1. $x > y$
2. $\frac{xy}{100} < x-y$

Algebra Percents

We are given that the rent collected increased by $x\%$ from 1997 to 1998 and decreased by $y\%$ from 1998 to 1999. Therefore, if the rent collected in 1997 was $\$R$, then the rent collected in 1998 was $(\$R)\left(1+\frac{x}{100}\right)$ and the rent

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Therefore, if the rent collected in 1997 was $\$R$, then the rent collected in 1998 was $(\$R)\left(1+\frac{x}{100}\right)$ and the rent collected in 1999 was $(\$R)\left(1+\frac{x}{100}\right)\left(1-\frac{y}{100}\right)$. Expanding this last expression, the rent collected in 1999 was $(\$R)\left(1+\frac{x-y}{100}-\frac{xy}{10{,}0000}\right)$, which is greater than $\$R$—the rent collected in 1997—if and only if $\frac{x-y}{100}-\frac{xy}{10{,}000}>0$.

1. Given that $x>y$, we have $\frac{x-y}{100}>0$, but it is not possible to determine whether $\frac{x-y}{100}-\frac{xy}{10{,}000}>0$. For example, if $x=20$ and $y=10$, then $x>y$ and $\frac{x-y}{100}-\frac{xy}{10{,}000}=\frac{10}{100}-\frac{200}{10{,}000}>0$. On the other hand, if $x=200$ and $y=100$, then $x>y$ and $\frac{x-y}{100}-\frac{xy}{10{,}000}=\frac{100}{100}-\frac{20{,}000}{10{,}000}<0$; NOT sufficient.
2. Given that $\frac{xy}{100}<x-y$, it follows that $\frac{xy}{10{,}000}<\frac{x-y}{100}$, and hence $\frac{x-y}{100}-\frac{xy}{10{,}000}>0$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

During an experiment, some water was removed from each of 6 water tanks. If the standard deviation of the volumes of water in the tanks at the beginning of the experiment was 10 gallons, what was the standard deviation of the volumes of water in the tanks at the end of the experiment?

1. For each tank, 30 percent of the volume of water that was in the tank at the beginning of the experiment was removed during the experiment.
2. The average (arithmetic mean) volume of water in the tanks at the end of the experiment was 63 gallons.

Algebra Statistics

Let $T_1, T_2, \ldots, T_6$ be the 6 original volumes of water at the beginning of the experiment and let T_{ave} be the average of the original volumes of water. We are given that the standard deviation of the original volumes of water is 10, namely $\sqrt{\frac{(T_1-T_{\text{ave}})^2+(T_2-T_{\text{ave}})^2+\ldots+(T_6-T_{\text{ave}})^2}{6}}=10$, and are to determine the standard deviation of the 6 new volumes of water at the end of the experiment. Note that one possibility for the 6 original volumes of water is 70, 70, 70, 90, 90, 90, since for these numbers we have $T_{\text{ave}}=80$, and hence $(T_1-T_{\text{ave}})^2+(T_2-T_{\text{ave}})^2+\ldots+(T_6-T_{\text{ave}})^2=10^2+10^2+\ldots+10^2=600$, so the standard deviation is $\sqrt{\frac{600}{6}}=\sqrt{100}=10$.

1. We are given that 30% was removed from each of the 6 tanks, so the 6 new volumes of water are $0.7T_1, 0.7T_2, \ldots, 0.7T_6$. The average of the new volumes of water is 0.7 times the average of the original volumes of water because $\frac{0.7T_1+0.7T_2+\ldots+0.7T_6}{6}=\frac{0.7(T_1+T_2+\ldots+T_6)}{6}=0.7T_{ave}$.

   Therefore, the standard deviation of the 6 new volumes of water is

$$\sqrt{\frac{(0.7T_1-0.7T_{ave})^2+(0.7T_2-0.7T_{ave})^2+\ldots+(0.7T_6-0.7T_{ave})^2}{6}}=$$

$$\sqrt{\frac{(0.7)^2(T_1-T_{ave})^2+(0.7)^2(T_2-T_{ave})^2+\ldots+(0.7)^2(T_6-T_{ave})^2}{6}}=$$

$$\sqrt{\frac{(0.7)^2[(T_1-T_{ave})^2+(T_2-T_{ave})^2+\ldots+(T_6-T_{ave})^2]}{6}}=$$

SUFFICIENT.

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$$0.7\sqrt{\frac{(T_1 - T_{ave})^2 + (T_2 - T_{ave})^2 + \ldots + (T_6 - T_{ave})^2}{6}} = (0.7)(10);$$

2. Given that the average of the 6 new volumes of water is 63, it is not possible to determine the standard deviation of the 6 new volumes of water. For example, if the original volumes of water were 70, 70, 70, 90, 90, 90 and each of the new volumes of water was 63, then the standard deviation of the original volumes of water would be 10, the average of the new volumes of water would be 63, and the standard deviation of the new volumes of water would be 0. On the other hand, if the original volumes of water were 70, 70, 70, 90, 90, 90 and the new volumes of water were 60, 60, 60, 66, 66, 66, then the standard deviation of the original volumes of water would be 10, the average of the new volumes of water would be 63, and the standard deviation of the new volumes of water would be greater than 0; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

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